

GONZALO CRUZ GÓMEZ

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Data Science & Engineering undergraduate (final year, URJC) with a background in Telecommunications Engineering. Builds production-ready data systems: end-to-end ML pipelines, distributed computing, and deep learning. Founder of Asociación Datalab. Seeking an internship in data science or data engineering.

EDUCATION

B.S. Data Science & Engineering	URJC	<i>2024 – Present</i>
• Final year (expected graduation 2027). Completed two academic years in one (2024–2025). Coursework: ML, Distributed Systems, Deep Learning, HPC, Statistical Inference.		
B.S. Telecommunications Engineering	UC3M	<i>2019 – 2023</i>
• 120+ credits in Networking, Physics and Signal Processing. Transferred to specialize in Data Science.		
Certificate of Advanced English (C2)	Cambridge Assessment	<i>2019</i>

PROJECTS

TripAdvisor Restaurants Data Pipeline	<i>Python, Apache Airflow, PySpark, Kafka, FastAPI, React</i>
• Orchestrated a 5-task Airflow DAG (ETL + ML + streaming) over 1M+ rows in 50k-row chunks: extract, clean, EDA, preprocess (normalization, OHE/label encoding by cardinality, incremental PCA), stream to Kafka.	
• Applied PySpark ML KMeans clustering with automatic K selection via silhouette scoring ($K = 10-80$); OHE or label encoding per column based on cardinality; all parameters in <code>config.toml</code> .	
• Built dual Spark Structured Streaming consumers: a fast filter path and an ML-scored path computing centroid distances and writing timestamped classifications in real time.	
• Developed a full-stack web app (FastAPI + React/Vite): restaurant search via cosine similarity within clusters; live scoring tab pushing KMeans classifications to the browser via Server-Sent Events.	
Forest Fire Spread Prediction	<i>R, GLMs, GAMs, Statistical Modeling</i>
• Built GLM and GAM regression models to predict fire-burned area from meteorological covariates; GAMs outperformed GLMs on held-out data with lower residual variance.	
• Resolved severe data quality issues (high skew, missing values, near-zero-variance predictors) via log transformation, imputation, and feature selection.	
• Applied cross-validation to compare model fits; selected GAMs as the final model based on lower RMSE and better capture of non-linear covariate effects.	
Stroke Risk Classification	<i>Python, Scikit-learn, TensorFlow, ANN</i>
• Trained an MLP on an imbalanced medical dataset, applying SMOTE and optimizing the decision threshold to minimize false negatives given the clinical cost of missed strokes.	
• Selected AUC-ROC and F1 as evaluation metrics over accuracy; tuned architecture and hyperparameters accordingly.	
• Benchmarked the MLP against a logistic regression baseline; the non-linear model improved recall on the stroke class while maintaining acceptable precision.	

EXPERIENCE

Founder & President — Asociación Datalab	<i>May 2025 – Present</i>
Established the university's first Data Science student association: defined the strategic roadmap, handled legal registration, and organized technical workshops.	
Private Tutor — Mathematics & Physics	<i>Sep 2021 – Present</i>
Mentored high-school students over 5 years, achieving a 100% pass rate and helping 90% of students improve by at least 3 grade points through personalized plans.	

TECHNICAL SKILLS

Programming:	Python (Advanced), R, SQL, C
Data Science:	Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Hadoop, Spark, Matplotlib, Excel
Engineering:	Apache Airflow, Kafka, Docker, Git/GitHub, Linux (Bash), Claude Code (agentic coding)
Languages:	Spanish (Native), English (C2), German (B1), Chinese (HSK3)